# Process & Decision Documentation

## Project/Assignment Decisions

For this week's sidequest, I used Week 6 example 5 code and added a change in background, characters and tilesheet. I changed the game visuals to make it look like night time.

Unfortunately, I couldn't figure out how to add sounds properly before the deadline, so in a way, my assignment decision is submitting without.

## Role-Based Process Evidence

Name: Queene Calading

Role(s): Designer

Primary responsibility for this work: Implementing change in asset visuals

Goal of Work Session:

- Find and incorporate the new assets into current playable code

Tools, Resources, or Inputs Used:

ChatGPT 5.2 ( Code help, debugging)

VSCode

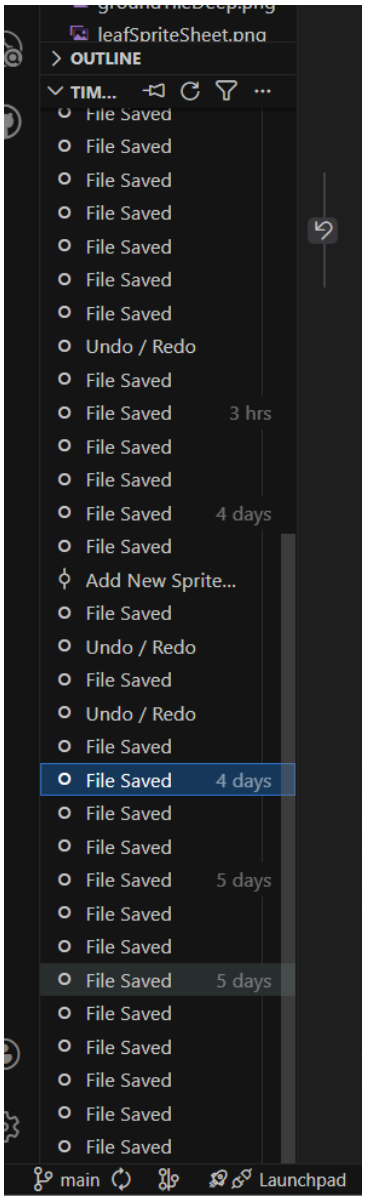
Github

Prior starter code given (W6 Example 5)

Itch.io (for assets)

```
// ---- images ----
// IMPORTANT:
// loadImage() is "preload-safe" only if p5 is actually tracking it inside preload()
// To make this robust even if your boot flow uses async/await, we wrap loadImage in
const playerImg = await loadImageAsync("assets/PinkMonSheet.png");
const boarImg = await loadImageAsync("assets/boarSpriteSheet.png");
const leafImg = await loadImageAsync("assets/leafSpriteSheet.png");
const boarImg = await loadImageAsync("assets/OwletSpritesheet.png");      You, 29 mi
const leafImg = await loadImageAsync("assets/BlobSprite3.png");
const fireImg = await loadImageAsync("assets/fireSpriteSheet.png");

const groundTileImg = await loadImageAsync("assets/groundTile.png");
const groundTileDeepImg = await loadImageAsync("assets/groundTileDeep.png");
const platformLCImg = await loadImageAsync("assets/platformLC.png");
```



## GenAI Documentation

**Date Used:** March 12, 2026

**Tool Disclosure:** ChatGPT 5.2

**Purpose of Use:** Coding guidance in how to change assets into the code, and debugging sizing issues

**Summary of Interaction**:

ChatGPT helped by providing step-by-step instructions in implementing the new assets I had found for the background, enemy, etc. It also helped in debugging my issues with the player character. It was still off and wrong at the end

**Human Decision Point(s):**

- Accepted debugging guidance and structural code suggestions, but adjusted parameters to match my pixel sizes and offsets. Rejected other suggestions in debugging after getting help from a friend that the only main issue was that we added a padding of 1 on our spritesheet when having 0 did the fix.

**Integrity & Verification Note:**

I verified that the given ideas were compatible with p5.js physics and drawing by checking the online p5.js library of what each parameter means in a line of code, and ensured no AI-generated code was copied without understanding (I asked questions and made it so that I was using AI to show me how / what I could do, rather than telling it what needs to be done and copy paste). I also adjusted values based on playtesting results

**Scope of GenAI Use:**

GenAI contributed to debugging guidance and example implementations, but didn't just make the code for me. Implementations to the actual code, with edits are done by me.

**Limitations or Misfires:**

- Some AI code examples included unnecessary complexity for my skill level, so I simplified them to maintain readability and ensure I fully understood each mechanic
- It couldn't debug what was wrong with my character and just kept jittering in its container / would be cut off (no feet).

## Appendix

### Transcript

https://chatgpt.com/share/69b44ae3-2fe4-8004-9617-89d6b84c8ecc

**Prompt:**
(Inserted the example 6 code as seen in link above)

Here is the side quest for this week (ignore the one for Week 6 in the Side Quest assignment outline): Find a free sprite sheet online (make sure to provide credit in your README) and change the characters for the player, leaf, and enemy. Find a free tile set online (make sure to provide credit in your README) and change the look of the level. Add sounds for jumping, attacking, taking damage, and collecting objects, and a looping background track. How do I go about changing the code to incorporate my own assets

**Response:**

Your codebase is already **very modular**, which is good — you mostly **won't change gameplay code**. You'll mainly update:

1. **Asset loading**
2. **Sprite/animation definitions**
3. **Tile graphics**
4. **Sound loading + event wiring**
5. **Parallax/background assets (optional)**

I'll walk through exactly **where in your architecture to change things**.

---

# 1. Replace Player / Enemy / Leaf Sprite Sheets

Your entities are created inside:

- `src/entities/PlayerEntity.js`
- `src/world/BoarSystem.js`

- `src/world/TileBuilder.js` (leafs)

The **sprites themselves come from `assets`**, which are loaded in:

src/AssetLoader.js

So **the first step is replacing assets in the loader.**

---

# 2. Edit `AssetLoader.js`

Your loader probably looks something like this (simplified):

```
assets.player = loadSpriteSheet("assets/player.png", {
 frameW: 32,
 frameH: 32
});
```

To use a new sprite sheet:

### Example

Suppose your sheet is:

assets/player_knight.png

Each frame:

64 x 64

Update:

```
assets.player = loadSpriteSheet("assets/player_knight.png", {
 frameW: 64,
 frameH: 64
});
```

---

# 3. Update Animations

Your **PlayerEntity** probably has something like:

```
this.sprite.addAni("idle", assets.player, 0, 3);
```

```
this.sprite.addAni("run", assets.player, 4, 9);
this.sprite.addAni("jump", assets.player, 10, 13);
this.sprite.addAni("attack", assets.player, 14, 17);
```

You must update these to match your new sheet layout.

Example:

```
this.sprite.addAni("idle", assets.player, 0, 5);
this.sprite.addAni("run", assets.player, 6, 11);
this.sprite.addAni("jump", assets.player, 12, 15);
this.sprite.addAni("attack", assets.player, 16, 21);
```

---

# 4. Replace Enemy (Boar)

Boars are built in:

src/world/BoarSystem.js

Look for something like:

```
e.addAni("walk", assets.boar, 0, 5);
e.addAni("death", assets.boar, 6, 10);
```

Replace with your enemy sprite:

assets/slime.png

Example:

```
e.addAni("walk", assets.slime, 0, 3);
e.addAni("death", assets.slime, 4, 7);
```

And update loader:

```
assets.slime = loadSpriteSheet("assets/slime.png", {
 frameW: 32,
 frameH: 32
});
```

---

# 5. Replace Leaf Collectible

Leafs come from **TileBuilder**.

Find something like:

```
this.leaf = new Group();
this.leaf.img = assets.leaf;
```

Then update the asset in `AssetLoader.js`:

```
assets.leaf = loadImage("assets/gem.png");
```

or animated collectible:

```
assets.gem = loadSpriteSheet("assets/gem.png", {
 frameW: 16,
 frameH: 16
});
```

---

# 6. Replace Tileset

Tiles are defined in:

data/levels.json

Your tile characters map to sprites in:

src/world/TileBuilder.js

Example:

```
this.ground.img = assets.ground;
this.platformsL.img = assets.platform;
```

Change assets:

```
assets.ground = loadImage("assets/tiles/grass.png");
assets.platform = loadImage("assets/tiles/platform.png");
assets.wall = loadImage("assets/tiles/rock.png");
```

Your **tile size must match your level config**:

From `levels.json`:

```
tileW: 24
tileH: 24
```

If your tileset uses **16x16** or **32x32**, you must update:

tileW

tileH

in `levels.json`.

---

# 7. Add Sounds

You already have a **SoundManager**, which is great.

## Step 1: Load sounds

Inside `main.js` after creating `soundManager`:

```
soundManager.load("jump", "assets/audio/jump.wav");
soundManager.load("attack", "assets/audio/attack.wav");
soundManager.load("hurt", "assets/audio/hurt.wav");
soundManager.load("leaf", "assets/audio/collect.wav");
soundManager.load("bgm", "assets/audio/music.mp3");
```

---

## Step 2: Hook events

You already have these:

In `Game.js`:

```
this.events.on("leaf:collected", () => this.sound.play("leaf"));
this.events.on("player:damaged", () => this.sound.play("hurt"));
this.events.on("boar:damaged", () => this.sound.play("hit"));
this.events.on("level:won", () => this.sound.play("win"));
```

Add:

```
this.events.on("player:jump", () => this.sound.play("jump"));
this.events.on("player:attack", () => this.sound.play("attack"));
```

---

## Step 3: Emit events

Inside `PlayerController`:

Jump:

```
this.events.emit("player:jump");
```

Attack:

this.events.emit("player:attack");

---

# 8. Add Looping Background Music

In `main.js` after loading:

soundManager.sfx.bgm.setLoop(true);
soundManager.play("bgm");

Or create a music channel.

---

# 9. Add Asset Credits

In your **README.md**:

Example:

ASSET CREDITS

Player sprites:
Knight character by FooBar Studios
https://opengameart.org/knight-character

Enemy sprites:
Slime enemy by Jane Smith
https://itch.io/slime-pack

Tileset:
Fantasy Tileset by Pixel Frog
https://itch.io/fantasy-tiles

Sound effects:
Jump/Attack sounds from Kenney.nl
https://kenney.nl/assets/audio

Music:
Forest Theme by Eric Matyas
https://soundimage.org

---

# 10. Where You Actually Change Code

You will likely modify **only these files**:

```
src/AssetLoader.js      ← most changes
src/entities/PlayerEntity.js
src/world/BoarSystem.js
src/world/TileBuilder.js
main.js                 ← sound loading
PlayerController.js     ← emit jump/attack events
levels.json             ← tile size if tiles change
```

Everything else stays untouched.

---

# Recommended Free Asset Sites

Good sources for this assignment:

### Sprites

itch.io/assets
opengameart.org
kenney.nl
craftpix.net (free section)

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### Tilesets

kenney.nl
itch.io tilesets
opengameart

---

### Sounds

kenney.nl
freesound.org
opengameart

---

# If you want, I can also show you

- **3 really good sprite packs that fit your game immediately**
- **a perfect tileset for your tilemap**
- **exact animation frame setups**

So you can plug them in **in under 20 minutes.**

my player is cut off, where the body is on top but head is bottom

Ah, got it — that sounds like a **sprite/animation offset problem** rather than a game logic issue. In p5play (or similar sprite systems), the “body upside-down” look usually happens when the **sprite’s origin/anchor** is misaligned. The sprite might be drawing its top-left corner at the player’s position, or the animation frame might be flipped vertically.

Here’s how to troubleshoot:

---

## 1. Check the sprite’s `ani` / image

- In `PlayerEntity.buildSprites()`, make sure the sprite images are **upright** and not accidentally flipped vertically.
- Example:

this.sprite.mirrorY(false); // ensure not flipped

---

## 2. Verify sprite’s `ani` anchor/offset

- p5play by default positions a sprite by its **center**, but some custom builds or loaded images might use **top-left**.
- If your sprite image’s “head” is at the bottom of the image file, you’ll see that cut-off effect.

You can adjust the origin:

this.sprite.ani.offsetY = -this.sprite.height/2; // shift sprite up so head aligns

Or explicitly set `center` mode:

this.sprite.center = createVector(this.sprite.width/2, this.sprite.height/2);

---

## 3. Check player controller position updates

- In `PlayerController.update()`, make sure `sprite.x` and `sprite.y` are **the sprite's center**, not bottom or top.
- Sometimes collision handling or physics moves the sprite by its top-left, while the image draws from center → misalignment.

---

## 4. Temporary debugging

- Draw a small dot at `sprite.x, sprite.y` to see where the engine thinks the player is:

```
fill('red');
circle(this.playerCtrl.sprite.x, this.playerCtrl.sprite.y, 4);
```

- If the red dot is near the head, your sprite's image anchor is reversed.

---

If you want, I can write a **quick patch for your `PlayerEntity`** so the sprite body aligns correctly with the physics/colliders and stops being upside-down.

Do you want me to do that?

yes

Perfect! Based on your `Level.js` and `PlayerEntity.js` setup, the easiest fix is to **ensure the sprite's origin is centered**, and any "y offset" is adjusted so the head sits at the top as expected. Here's a patch you can apply inside your `PlayerEntity.buildSprites()` method:

```
buildSprites() {
 // assuming you already create sprite(s) here
 this.sprite = createSprite(this.x, this.y);

 // attach animations
 for (const [name, aniFrames] of Object.entries(this.anis)) {
   this.sprite.addAnimation(name, ...aniFrames);
 }

 // --- FIX ORIGIN / OFFSET ---
 // 1. Make sprite draw from center
 this.sprite.setCollider("rectangle", 0, 0, this.sprite.width, this.sprite.height);
 this.sprite.mirrorY(false);      // ensure no vertical flip
```

```
 this.sprite.rotation = 0;        // just in case

 // 2. Optional: adjust vertical offset if head seems below body
 // Use half-height shift to align correctly
 this.sprite.offset.y = -this.sprite.height / 2;

 // 3. Ensure physics/collisions use center
 this.sprite.collider = "rectangle";
}
```

---

## Notes:

1. `this.sprite.offset.y` moves the image relative to the sprite's actual physics/collider.
    - Negative values shift it **up**, so the head appears at the top.
    - Positive values shift **down**.
2. Make sure `mirrorY(false)` is called if the sprite was flipped vertically anywhere.
3. Collider should match sprite dimensions; otherwise, collisions may feel off.

---

💡 **Quick debug trick:**
Draw the sprite's center with a small red circle:

```
fill('red');
circle(this.sprite.x, this.sprite.y, 4);
```

If the red dot is at the middle of the player body, your sprite is now correctly aligned.

---

If you want, I can also write a **full drop-in replacement for `PlayerEntity.buildSprites()`** that fixes both alignment **and attack hitbox offsets** so the player is fully upright and collides properly.

Do you want me to do that?

**Prompt:**

buildSprites() { const { frameW = 32, frameH = 32 } = this.tilesCfg; this.sprite = new Sprite(this.startX, this.startY, frameW, frameH); this.sprite.rotationLock = true; const anis = this.assets?.playerAnis; const img = this.assets?.playerImg; if (img) this.sprite.spriteSheet = img; if (anis && typeof anis === "object") { this.sprite.anis.w = frameW; this.sprite.anis.h = frameH; this.sprite.anis.offset.y = this.ANI_OFFSET_Y; this.sprite.addAnis(anis); this._setAni("idle"); } else { this.sprite.img = img; } // collider this.sprite.w =

this.COLLIDER_W; this.sprite.h = this.COLLIDER_H; this.sprite.friction = 0; this.sprite.bounciness = 0; // ground sensor (query-only) this.sensor = new Sprite(); this.sensor.x = this.sprite.x; this.sensor.y = this.sprite.y + this.sprite.h / 2; this.sensor.w = this.sprite.w; this.sensor.h = 2; this.sensor.mass = 0.01; this.sensor.removeColliders(); this.sensor.visible = false; const j = new GlueJoint(this.sprite, this.sensor); j.visible = false; return this; } reset() { this.health = this.maxHealth; this.dead = false; this.pendingDeath = false; this.deathAnimStarted = false; this.invulnTimer = 0; this.knockTimer = 0; this.attacking = false; this.attackFrameCounter = 0; this.attackHitThisSwing = false; if (!this.sprite) return; this.sprite.x = this.startX; this.sprite.y = this.startY; this.sprite.vel.x = 0; this.sprite.vel.y = 0; this.sprite.tint = "#ffffff"; this._setAni("idle"); } // ---------- // queries // ---------- isGrounded(solids) { const s = this.sensor; if (!s) return false; // Allow either { ground, platformsL, ... } OR an array [ground, platformsL, ...] const list = Array.isArray(solids) ? solids : Object.values(solids || {}); for (const g of list) { if (g && s.overlapping(g)) return true; } return false; } // ---------- // timers // ---------- tickTimers() { if (this.invulnTimer > 0) this.invulnTimer--; if (this.knockTimer > 0) this.knockTimer--; } // ---------- // actions (mutations) // ---------- stopX() { this.sprite.vel.x = 0; } moveLeft() { this.sprite.vel.x = -this.MOVE_SPEED; this.sprite.mirror.x = true; } moveRight() { this.sprite.vel.x = this.MOVE_SPEED; this.sprite.mirror.x = false; } jump() { this.sprite.vel.y = -1 * this.JUMP_STRENGTH; } startAttack() { this.attacking = true; this.attackHitThisSwing = false; this.attackFrameCounter = 0; this.stopX(); this._playAni("attack", 0); } markAttackHit() { this.attackHitThisSwing = true; } clampToBounds(bounds) { const half = (this.sprite?.w ?? 0) / 2; const maxX = (bounds?.levelW ?? this.sprite.x) - half; this.sprite.x = constrain(this.sprite.x, half, maxX); } // ---------- // animation (visual state) // ---------- applyAnimation({ grounded, won }) { if (!this.sprite?.anis || Object.keys(this.sprite.anis).length === 0) return; // ---- DEAD: play once, then hold last frame (prevents infinite loop) if (this.dead) { if (!this.deathAnimStarted) { this.deathAnimStarted = true; this._playAni("death", 0); // If available, disable looping this.sprite.ani?.noLoop?.(); } else { this._setAni("death"); // Hard clamp to last frame (works even if noLoop doesn't exist) const def = this.assets?.playerAnis?.death; const frames = Number(def?.frames ?? 1); if (this.sprite.ani) this.sprite.ani.frame = Math.max(0, frames - 1); } return; } if (won) { // optional: choose idle when winning this._setAni("idle"); return; } if (this.knockTimer > 0 || this.pendingDeath) { this._setAniFrame("hurtPose", 1); return; } // attack animation is started by startAttack(); don't stomp it here if (this.attacking) return; if (!grounded) { const f = this.sprite.vel.y < 0 ? 0 : 1; this._setAniFrame("jump", f); return; } const moving = Math.abs(this.sprite.vel.x) > 0.01; this._setAni(moving ? "run" : "idle"); } // ---------- // damage / effects // ---------- takeDamageFromX(sourceX) { if (this.invulnTimer > 0 || this.dead) return false; this.health = Math.max(0, this.health - 1); if (this.health <= 0) this.pendingDeath = true; this.invulnTimer = this.INVULN_FRAMES; this.knockTimer = this.KNOCK_FRAMES; const dir = this.sprite.x < sourceX ? -1 : 1; this.sprite.vel.x = dir * this.KNOCKBACK_X; this.sprite.vel.y = -this.KNOCKBACK_Y; // cancel attack this.attacking = false; this.attackFrameCounter = 0; this.attackHitThisSwing = false; return true; } applyHurtBlinkTint() { if (!this.sprite) return; if (!this.dead && this.invulnTimer > 0) { this.sprite.tint = Math.floor(this.invulnTimer / 4) % 2 === 0 ? "#ff5050" : "#ffffff"; } else { this.sprite.tint = "#ffffff"; } } }

**Response:**

Ah! Thanks, now I can see exactly how your `PlayerEntity` is structured. The issue where the **body appears on top and the head is at the bottom** is almost certainly because of **`this.ANI_OFFSET_Y`** and the sprite frame origin.

In your current code:

this.sprite.anis.offset.y = this.ANI_OFFSET_Y;

That offset determines **how the animation frames are drawn relative to the sprite's position**. If `ANI_OFFSET_Y` is positive, it will push the sprite **down**, which can make it look like the body is above the head.

---

## Quick Fix

1. Make `ANI_OFFSET_Y` **negative**, so the sprite shifts **up** to align the head at the top:

```
this.ANI_OFFSET_Y = - (frameH / 2); // move the sprite up by half its height
```

2. Or, if you want fine control, you can tweak it until it looks right:

```
this.sprite.anis.offset.y = -4; // trial and error
```

3. Optional: make sure the collider matches the visible sprite:

```
this.sprite.w = this.COLLIDER_W;
this.sprite.h = this.COLLIDER_H;
// Center the collider on the sprite
this.sprite.offset.x = 0;
this.sprite.offset.y = 0;
```

---

## Debug Tip

Draw the sprite's origin as a dot to see where it's anchored:

```
fill("red");
circle(this.sprite.x, this.sprite.y, 4);
```

That will help you adjust `ANI_OFFSET_Y` so the head lines up correctly.

---

If you want, I can rewrite your **`buildSprites()`** method so the sprite is perfectly upright, the collider is aligned, and the ground sensor works—basically a drop-in version that fixes the "head below body" issue.

Do you want me to do that?

**Prompt:**
fixed the sprite idle but when I run or jump, I can only see half the body and top frame of sprite container shows part of bottom half of body

**Response:**

This is a **sprite sheet frame clipping issue**, not a physics or collider problem. Your idle works because that animation probably has the correct frame size, but the **run/jump animations are taller than the frame size you're telling p5play**.

Right now you force every animation frame to:

```
frameW = 32
frameH = 32
```

But if your run/jump sprites are **taller than 32px**, p5play will **crop the image**, which is why:

- you see **half the body**
- the **top of the frame shows the bottom of the sprite**

---

# Fix 1 (Most Likely): Increase frame height

If your sprite frames are actually **48px or 64px tall**, change:

```
const { frameW = 32, frameH = 32 } = this.tilesCfg;
```

to something like:

```
const { frameW = 32, frameH = 48 } = this.tilesCfg;
```

or

```
const { frameW = 32, frameH = 64 } = this.tilesCfg;
```

Then keep:

```
this.sprite.anis.w = frameW;
this.sprite.anis.h = frameH;
```

---

# Fix 2: Different frame size for animations

If idle is 32x32 but run/jump are larger, set the **largest size**:

```
this.sprite.anis.w = 32;
this.sprite.anis.h = 48;
```

p5play needs the **full frame size** or it will crop.

---

## Fix 3 (If frames are centered differently)

After fixing height, you may need a small offset tweak:

this.sprite.anis.offset.y = -8;

But **do this only after fixing frame height**.

---

## Quick way to confirm the problem

Open the sprite sheet and check:

- width of **one frame**
- height of **one frame**

Example:

| Animation | Frame Size |
|---|---|
| idle | 32x32 |
| run | **32x48** |
| jump | **32x48** |

If that's the case, your game must use **32x48**, not 32x32.

---

✅ **90% chance your fix is simply:**

const { frameW = 32, frameH = 48 } = this.tilesCfg;

---

If you want, send **one screenshot of the sprite sheet**, and I can tell you the **exact frame width/height** you should use so the clipping disappears.

Prompt:
Uploaded image

Response:
Thanks for the sprite sheet — that helps a lot. 👍

Your image size is **258 × 204 px**. That strongly suggests the frames are arranged like this:

- **8 columns**
- **6 rows**
- Each frame ≈ **32 × 34 px** (not 32 × 32)

Why:

- 8 × 32 ≈ **256** (matches the width with a tiny margin)
- 6 × 34 = **204** (exactly your height)

So the **actual frame height is 34**, not 32. When you set:

```
this.sprite.anis.w = 32;
this.sprite.anis.h = 32;
```

p5play crops each frame to **32px tall**, which cuts off the bottom of the character during **run/jump**, making the body look split.

---

## ✅ Fix

Change your animation frame height to **34**.

```
this.sprite.anis.w = 32;
this.sprite.anis.h = 34;
```

Keep the collider the same if you want tighter physics:

```
this.sprite.w = this.COLLIDER_W;
this.sprite.h = this.COLLIDER_H;
```

---

# Optional: adjust vertical alignment

Because the sprite is now slightly taller, you may want a small offset:

```
this.sprite.anis.offset.y = -1;
```

or

```
this.sprite.anis.offset.y = -2;
```

Just tweak until the feet sit correctly on the ground.

---

# Why idle looked fine

Idle probably looked correct because that animation frame didn't extend to the very bottom of the 34px cell, so the 32px crop wasn't obvious.

Run/jump extend further downward → **visible cropping**.

---

✅ **Final working block:**

```
this.sprite.anis.w = 32;
this.sprite.anis.h = 34;
this.sprite.anis.offset.y = -2;
this.sprite.addAnis(anis);
this._setAni("idle");
```

---

💡 If you'd like, I can also show you a **much cleaner way to load this sheet in p5play** so you **never have to guess frame sizes again**, and it will automatically detect the grid

**Prompt:**

// src/entities/PlayerEntity.js // Player entity (WORLD entity). // // Responsibilities: // - Own player state (health/timers/flags) // - Own player sprites (main sprite + ground sensor) // - Provide small action methods (move/jump/attack/damage) // - Apply animation based on current state (visual state driven by world state) // // Non-goals: // - Does NOT read input directly (PlayerController does) // - Does NOT decide game win state (Level does) export class PlayerEntity { constructor(pkg, assets) { this.pkg = pkg; this.assets = assets; this.tuning = pkg.tuning || {}; this.tilesCfg = pkg.tiles || {}; this.bounds = pkg.bounds || {}; this.levelData = pkg.level || {}; // sprites this.sprite = null; this.sensor = null; // spawn const ps = this.levelData.playerStart || { x: this.tilesCfg.frameW ?? 32, y: (this.bounds.levelH ?? 0) - (this.tilesCfg.tileH ?? 24) * 4, }; this.startX = ps.x; this.startY = ps.y; // stats this.maxHealth = Number(this.tuning.player?.maxHealth ?? 3); this.health = this.maxHealth; // state this.dead = false; this.pendingDeath = false; // death animation latch (prevents looping forever) this.deathAnimStarted = false; // timers this.invulnTimer = 0; this.knockTimer = 0; // attack this.attacking = false; this.attackFrameCounter = 0; this.attackHitThisSwing = false; // tuning defaults this.MOVE_SPEED = Number(this.tuning.player?.moveSpeed ?? 1.5); this.JUMP_STRENGTH = Number(this.tuning.player?.jumpStrength ?? 4.5); this.INVULN_FRAMES = Number(this.tuning.player?.invulnFrames ?? 45); this.KNOCK_FRAMES = Number(this.tuning.player?.knockFrames ?? 30); this.KNOCKBACK_X = Number(this.tuning.player?.knockbackX ?? 2.0); this.KNOCKBACK_Y = Number(this.tuning.player?.knockbackY ?? 3.2); this.COLLIDER_W = Number(this.tuning.player?.w ?? 18); this.COLLIDER_H = Number(this.tuning.player?.h ?? 12); this.ANI_OFFSET_Y =

Number(this.tuning.player?.aniOffsetY ?? -8); // attack window this.ATTACK_START = Number(this.tuning.player?.attackStartFrame ?? 4); this.ATTACK_END = Number(this.tuning.player?.attackEndFrame ?? 8); this.ATTACK_FINISH = Number(this.tuning.player?.attackFinishFrame ?? 12); } // ---------- // animation safety // ---------- _hasAni(name) { return !!(this.sprite?.anis && this.sprite.anis[name]); } _setAni(name) { if (!this._hasAni(name)) return false; this.sprite.ani = name; return true; } _setAniFrame(name, frame) { if (!this._setAni(name)) return false; if (this.sprite.ani) this.sprite.ani.frame = frame; return true; } _playAni(name, startFrame = 0) { if (!this._setAni(name)) return false; if (this.sprite.ani) { this.sprite.ani.frame = startFrame; this.sprite.ani.play?.(); } return true; } // ---------- // build / reset // ---------- buildSprites() { const { frameW = 30, frameH = 64 } = this.tilesCfg; this.sprite = new Sprite(this.startX, this.startY, frameW, frameH); this.sprite.rotationLock = true; const anis = this.assets?.playerAnis; const img = this.assets?.playerImg; if (img) this.sprite.spriteSheet = img; if (anis && typeof anis === "object") { this.sprite.anis.w = frameW; this.sprite.anis.h = frameH; this.sprite.anis.offset.y = -9; this.sprite.addAnis(anis); this._setAni("idle"); } else { this.sprite.img = img; } // collider this.sprite.w = this.COLLIDER_W; this.sprite.h = this.COLLIDER_H; this.sprite.friction = 0; this.sprite.bounciness = 0; // ground sensor (query-only) this.sensor = new Sprite(); this.sensor.x = this.sprite.x; this.sensor.y = this.sprite.y + this.sprite.h / 2; this.sensor.w = this.sprite.w; this.sensor.h = 2; this.sensor.mass = 0.01; this.sensor.removeColliders(); this.sensor.visible = false; const j = new GlueJoint(this.sprite, this.sensor); j.visible = false; return this; } reset() { this.health = this.maxHealth; this.dead = false; this.pendingDeath = false; this.deathAnimStarted = false; this.invulnTimer = 0; this.knockTimer = 0; this.attacking = false; this.attackFrameCounter = 0; this.attackHitThisSwing = false; if (!this.sprite) return; this.sprite.x = this.startX; this.sprite.y = this.startY; this.sprite.vel.x = 0; this.sprite.vel.y = 0; this.sprite.tint = "#ffffff"; this._setAni("idle"); } // ---------- // queries // ---------- isGrounded(solids) { const s = this.sensor; if (!s) return false; // Allow either { ground, platformsL, ... } OR an array [ground, platformsL, ...] const list = Array.isArray(solids) ? solids : Object.values(solids || {}); for (const g of list) { if (g && s.overlapping(g)) return true; } return false; } // ---------- // timers // ---------- tickTimers() { if (this.invulnTimer > 0) this.invulnTimer--; if (this.knockTimer > 0) this.knockTimer--; } // ---------- // actions (mutations) // ---------- stopX() { this.sprite.vel.x = 0; } moveLeft() { this.sprite.vel.x = -this.MOVE_SPEED; this.sprite.mirror.x = true; } moveRight() { this.sprite.vel.x = this.MOVE_SPEED; this.sprite.mirror.x = false; } jump() { this.sprite.vel.y = -1 * this.JUMP_STRENGTH; } startAttack() { this.attacking = true; this.attackHitThisSwing = false; this.attackFrameCounter = 0; this.stopX(); this._playAni("attack", 0); } markAttackHit() { this.attackHitThisSwing = true; } clampToBounds(bounds) { const half = (this.sprite?.w ?? 0) / 2; const maxX = (bounds?.levelW ?? this.sprite.x) - half; this.sprite.x = constrain(this.sprite.x, half, maxX); } // ---------- // animation (visual state) // ---------- applyAnimation({ grounded, won }) { if (!this.sprite?.anis || Object.keys(this.sprite.anis).length === 0) return; // ---- DEAD: play once, then hold last frame (prevents infinite loop) if (this.dead) { if (!this.deathAnimStarted) { this.deathAnimStarted = true; this._playAni("death", 0); // If available, disable looping this.sprite.ani?.noLoop?.(); } else { this._setAni("death"); // Hard clamp to last frame (works even if noLoop doesn't exist) const def = this.assets?.playerAnis?.death; const frames = Number(def?.frames ?? 1); if (this.sprite.ani) this.sprite.ani.frame = Math.max(0, frames - 1); } return; } if (won) { // optional: choose idle when winning this._setAni("idle"); return; } if (this.knockTimer > 0 || this.pendingDeath) { this._setAniFrame("hurtPose", 1); return; } // attack animation is started by startAttack(); don't stomp it here if (this.attacking) return; if (!grounded) { const f = this.sprite.vel.y < 0 ? 0 : 1; this._setAniFrame("jump", f); return; } const moving = Math.abs(this.sprite.vel.x) > 0.01; this._setAni(moving ? "run" : "idle"); } // ---------- // damage / effects // ---------- takeDamageFromX(sourceX) { if (this.invulnTimer > 0 || this.dead) return false; this.health = Math.max(0, this.health - 1); if (this.health <= 0) this.pendingDeath = true; this.invulnTimer = this.INVULN_FRAMES; this.knockTimer = this.KNOCK_FRAMES; const dir = this.sprite.x < sourceX ? -1 : 1; this.sprite.vel.x = dir * this.KNOCKBACK_X; this.sprite.vel.y = -this.KNOCKBACK_Y; // cancel attack

this.attacking = false; this.attackFrameCounter = 0; this.attackHitThisSwing = false; return true; } applyHurtBlinkTint() { if (!this.sprite) return; if (!this.dead && this.invulnTimer > 0) { this.sprite.tint = Math.floor(this.invulnTimer / 4) % 2 === 0 ? "#ff5050" : "#ffffff"; } else { this.sprite.tint = "#ffffff"; } } }

**Response:**

The problem is here in your `buildSprites()`:

const { frameW = 30, frameH = 64 } = this.tilesCfg;

But the sprite sheet you showed earlier uses **32 × 34 frames**.

Right now you are telling p5play each frame is **30 × 64**, which causes:

- wrong cropping
- half the body disappearing
- the next row of the sheet leaking into the frame

---

# ✅ Correct Fix

Change this line:

const { frameW = 30, frameH = 64 } = this.tilesCfg;

to:

const { frameW = 32, frameH = 34 } = this.tilesCfg;

---

# Then fix the animation setup

Replace this block:

this.sprite.anis.w = frameW;
this.sprite.anis.h = frameH;
this.sprite.anis.offset.y = -9;

with:

this.sprite.anis.w = 32;

```
this.sprite.anis.h = 34;
this.sprite.anis.offset.y = -2;
```

---

# Final working `buildSprites()` section

```
buildSprites() {
 const { frameW = 32, frameH = 34 } = this.tilesCfg;

 this.sprite = new Sprite(this.startX, this.startY, frameW, frameH);
 this.sprite.rotationLock = true;

 const anis = this.assets?.playerAnis;
 const img = this.assets?.playerImg;

 if (img) this.sprite.spriteSheet = img;

 if (anis && typeof anis === "object") {
   this.sprite.anis.w = 32;
   this.sprite.anis.h = 34;
   this.sprite.anis.offset.y = -2;

   this.sprite.addAnis(anis);
   this._setAni("idle");
 } else {
   this.sprite.img = img;
 }
```

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# Why this fixes everything

Your sprite sheet grid is:

| columns | rows | frame size |
|---|---|---|
| 8 | 6 | **32 × 34** |

Using the wrong frame size causes p5play to read frames like this:

[ half frame ]
[ next row ]

which is exactly the glitch you described.

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# Optional (recommended improvement)

Your collider is much smaller than the sprite:

```
this.COLLIDER_W = 18
this.COLLIDER_H = 12
```

That's fine, but you may want the sprite centered better:

```
this.sprite.anis.offset.y = -4;
```

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✅ If you want, I can also show you a **much cleaner p5play setup for sprite sheets** that will:

- auto-detect frames
- fix alignment automatically
- reduce ~10 lines of code here

and make future sprites easier to add.